

Lecture 10: 02/09/2026

Lecturer: Mauricio Daros

10.1 Matrix Power Series and the Matrix Exponential

The matrix exponential will provide the link between infinitesimal transition rates and the transition probabilities of a continuous-time Markov chain. We begin by collecting the matrix-function results that will be needed below.

Let $(c_k)_{k \geq 0} \subseteq \mathbb{R}$, and consider the complex power series

$$f(z) = \sum_{k=0}^{\infty} c_k z^k \quad (|z| < R),$$

where $R \in (0, \infty]$ is its radius of convergence. Thus the series converges absolutely for every $z \in \mathbb{C}$ with $|z| < R$.

Definition 10.1.1. The *spectral radius* of $A \in \mathbb{R}^{d \times d}$ is

$$\rho(A) := \max\{|\lambda| : \lambda \in \mathbb{C} \text{ is an eigenvalue of } A\}.$$

Lemma 10.1.1. If $A \in \mathbb{R}^{d \times d}$ satisfies $\rho(A) < R$, then

$$\sum_{k=0}^{\infty} c_k A^k$$

converges absolutely.

Proof. Choose r such that $\rho(A) < r < R$, and fix a submultiplicative matrix norm. By Gelfand's spectral-radius formula,

$$\lim_{k \rightarrow \infty} \|A^k\|^{1/k} = \rho(A);$$

see, for example, [HJ13]. Hence there is $N \geq 1$ such that $\|A^k\| \leq r^k$ for every $k \geq N$. Therefore,

$$\sum_{k=N}^{\infty} |c_k| \|A^k\| \leq \sum_{k=N}^{\infty} |c_k| r^k < \infty,$$

because the scalar power series converges absolutely at r . □

Definition 10.1.2. For $A \in \mathbb{R}^{d \times d}$ with $\rho(A) < R$, define

$$f(A) := \sum_{k=0}^{\infty} c_k A^k.$$

This is well-defined by Lemma 10.1.1.

The definition does not depend on the norm used to test convergence, since $\mathbb{R}^{d \times d}$ is finite-dimensional and all its norms induce the same topology.

Lemma 10.1.2. Suppose that $A \in \mathbb{R}^{d \times d}$ is diagonalizable over $\mathbb{C}$, with

$$A = UDU^{-1}, \quad U \in \mathbb{C}^{d \times d}, \quad D = \text{diag}(\lambda_1, \dots, \lambda_d).$$

If $\rho(A) < R$, then

$$f(A) = U \text{diag}(f(\lambda_1), \dots, f(\lambda_d))U^{-1}.$$

Proof. For every $N \geq 0$,

$$\sum_{k=0}^N c_k A^k = U \left(\sum_{k=0}^N c_k D^k \right) U^{-1}.$$

Since $|\lambda_i| \leq \rho(A) < R$ for every i , letting $N \rightarrow \infty$ gives the claimed identity. $\square$

Lemma 10.1.3. Let $A \in \mathbb{R}^{d \times d}$ satisfy $\rho(A) < R$, and let $\lambda_1, \dots, \lambda_d$ be its eigenvalues, counted with algebraic multiplicity. Then the eigenvalues of $f(A)$ are

$$f(\lambda_1), \dots, f(\lambda_d),$$

again counted with algebraic multiplicity.

Proof. Work over $\mathbb{C}$ and choose an invertible matrix U such that $T = U^{-1}AU$ is upper triangular with diagonal entries $\lambda_1, \dots, \lambda_d$. Set

$$S_N(A) := \sum_{k=0}^N c_k A^k.$$

Then

$$U^{-1}S_N(A)U = S_N(T) \longrightarrow U^{-1}f(A)U.$$

Each $S_N(T)$ is upper triangular, with i -th diagonal entry $\sum_{k=0}^N c_k \lambda_i^k$. Thus $U^{-1}f(A)U$ is upper triangular with diagonal entries $f(\lambda_1), \dots, f(\lambda_d)$, which are its eigenvalues. $\square$

Definition 10.1.3. The *matrix exponential* of $A \in \mathbb{R}^{d \times d}$ is

$$\exp(A) = e^A := \sum_{n=0}^{\infty} \frac{A^n}{n!}.$$

Since the scalar exponential series has infinite radius of convergence, this series converges absolutely for every matrix A .

Lemma 10.1.4. For every $A \in \mathbb{R}^{d \times d}$ and every $s, t \in \mathbb{R}$,

$$e^{0A} = \mathbf{I}, \quad e^{(s+t)A} = e^{sA} e^{tA}.$$

Proof. The first identity follows directly from the definition. Absolute convergence permits the Cauchy product

$$\begin{aligned} e^{sA} e^{tA} &= \sum_{k=0}^{\infty} \left(\sum_{m=0}^k \frac{s^m t^{k-m}}{m!(k-m)!} \right) A^k \\ &= \sum_{k=0}^{\infty} \frac{(s+t)^k}{k!} A^k = e^{(s+t)A}, \end{aligned}$$

where the second equality is the binomial theorem. $\square$

Lemma 10.1.5. Let $A \in \mathbb{R}^{d \times d}$. The function $t \mapsto e^{tA}$ is differentiable on $\mathbb{R}$, and

$$\frac{d}{dt} e^{tA} = A e^{tA} = e^{tA} A.$$

Proof. Fix $T > 0$ and a submultiplicative matrix norm. The estimates

$$\sum_{n=0}^{\infty} \sup_{|t| \leq T} \left\| \frac{t^n A^n}{n!} \right\| \leq e^{T\|A\|},$$

and the series of derivatives is bounded by

$$\sum_{n=1}^{\infty} \sup_{|t| \leq T} \left\| \frac{nt^{n-1} A^n}{n!} \right\| \leq \|A\| e^{T\|A\|}.$$

Both series therefore converge uniformly on $[-T, T]$, so term-by-term differentiation is valid. It gives

$$\frac{d}{dt} e^{tA} = \sum_{n=1}^{\infty} \frac{nt^{n-1} A^n}{n!} = A \sum_{n=0}^{\infty} \frac{t^n A^n}{n!} = A e^{tA}.$$

Since A commutes with each of its powers, $A e^{tA} = e^{tA} A$. □

Corollary 10.1.1. *The function $X(t) = e^{tA}$ is the unique differentiable matrix-valued solution of*

$$X'(t) = AX(t), \quad X(0) = \mathbf{I}.$$

Proof. Lemma 10.1.5 shows that e^{tA} solves the system and has the required initial value. If Y is any other solution, then

$$\frac{d}{dt} (e^{-tA} Y(t)) = -A e^{-tA} Y(t) + e^{-tA} A Y(t) = 0.$$

Hence $e^{-tA} Y(t) = Y(0) = \mathbf{I}$. The semigroup property gives $e^{-tA} e^{tA} = \mathbf{I}$, and therefore $Y(t) = e^{tA}$. □

Exercise 10.1.2. *Recall the definition of matrix inverse: A^{-1} denotes the matrix that satisfies $A^{-1}A = \mathbf{I}$. Let $f(z) = 1/(1 - z)$ for $|z| < 1$. Show that if $\rho(A) < 1$, then $f(A) = (\mathbf{I} - A)^{-1}$, that is, $f(A)$ satisfies $f(A)(\mathbf{I} - A) = \mathbf{I}$.*

Lemma 10.1.6. *Let f , g , and h be represented by power series about 0, with radii of convergence R_f , R_g , and R_h , respectively. Let $A \in \mathbb{R}^{d \times d}$. Suppose that there is an r such that*

$$\rho(A) < r < \min\{R_g, R_h\},$$

and, for every $|z| \leq r$,

$$|g(z)| < R_f,$$

while $h(z) = f(g(z))$ for every $|z| < r$. Then

$$f(g(A)) = h(A).$$

Proof. The series defining $g(A)$ and $h(A)$ converge by Lemma 10.1.1. By Lemma 10.1.3, every eigenvalue of $g(A)$ has the form $g(\lambda)$ for an eigenvalue λ of A . Hence $\rho(g(A)) < R_f$, so the series defining $f(g(A))$ also converges. Since $h = f \circ g$ on a neighbourhood of the eigenvalues of A , the composite-function theorem gives $f(g(A)) = h(A)$; see [Hig08, Theorem 1.17]. □

Definition 10.1.4. The *power-series matrix logarithm* of $A \in \mathbb{R}^{d \times d}$ is

$$\log A := \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (A - \mathbf{I})^n,$$

whenever $\rho(A - \mathbf{I}) < 1$. This is a local choice of the matrix logarithm; a matrix logarithm is not unique in general.

Lemma 10.1.7. *Let $A \in \mathbb{R}^{d \times d}$. If $\rho(A - \mathbf{I}) < 1$, then*

$$\exp(\log A) = A.$$

In the reverse direction, if $\rho(A) < \log 2$, then $\log(\exp A)$ is defined and

$$\log(\exp A) = A.$$

Proof. Set $B = A - \mathbf{I}$ and choose r such that $\rho(B) < r < 1$. Apply Lemma 10.1.6 at B with

$$f(z) = e^z, \quad g(z) = \log(1 + z), \quad h(z) = 1 + z.$$

Its hypotheses hold on $|z| \leq r$, and hence

$$\exp(\log A) = \exp(\log(\mathbf{I} + B)) = \mathbf{I} + B = A.$$

Now suppose that $\rho(A) < \log 2$, and choose r such that $\rho(A) < r < \log 2$. For $|z| \leq r$,

$$|e^z - 1| \leq e^{|z|} - 1 \leq e^r - 1 < 1.$$

Apply Lemma 10.1.6 at A with

$$f(z) = \log(1 + z), \quad g(z) = e^z - 1, \quad h(z) = z.$$

This also shows that the series defining $\log(\exp A)$ converges, and it gives $\log(\exp A) = h(A) = A$. $\square$

10.1.1 An application in discrete-time Markov Chains

This subsection is based on [DS84].

Suppose that $(X_n)_{n \in \mathbb{N}}$ is a Markov Chain on $\{1, \dots, d\}$ with transition matrix $\mathbf{P}$. Assume that d' states are absorbing ($d' < d$), and without generality, we assume that the absorbing states are $1, \dots, d'$. Assume that all the remaining states are transient. The matrix $\mathbf{P}$ assumes the form

$$\mathbf{P} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{R} & \mathbf{Q} \end{pmatrix} \quad (10.1)$$

for some matrices $\mathbf{R} \in \mathbb{R}^{(d-d') \times d'}$ and $\mathbf{Q} \in \mathbb{R}^{(d-d') \times (d-d')}$. Recall the definition

$$g_{ij} = \mathbb{E}(N_j \mid X_0 = i),$$

where N_j is the number of times the chain visits the state j . Let $\mathbf{G} = (g_{ij})_{i,j \in \{d'+1, \dots, d\}}$. Observe that

$$\mathbf{G} = \mathbf{I} + \mathbf{Q} + \mathbf{Q}^2 + \dots$$

See [Sap26, Definition 1.3.8]. One can check that $\rho(\mathbf{Q}) < 1$. Then, Exercise 10.1.2 implies that

$$\mathbf{G} = (\mathbf{I} - \mathbf{Q})^{-1}$$

For instance, consider the simple symmetric random walk on $\{1, 2, 3, 4\}$, where 1 and 4 are absorbing. Using the state ordering $(1, 4, 2, 3)$, so that the first two rows and columns correspond to the absorbing states, its transition matrix is

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}.$$

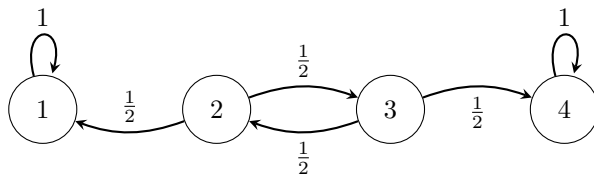


Figure 10.1: The simple symmetric random walk on $\{1, 2, 3, 4\}$, absorbed upon reaching state 1 or 4.

Then

$$\mathbf{Q} = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix},$$

and

$$(\mathbf{I} - \mathbf{Q})^{-1} = \begin{pmatrix} \frac{4}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{4}{3} \end{pmatrix}$$

Therefore, when the walk starts at state 2, it returns to that state about 1.33 times before it is absorbed.

Exercise 10.1.3. Suppose that $(X_n)_{n \in \mathbb{N}}$ is a Markov Chain on $\{1, \dots, d\}$ with transition matrix $\mathbf{P}$. Assume that d' states are absorbing ($d' < d$), and without generality, we assume that the absorbing states are $1, \dots, d'$. Assume that all the remaining states are transient. Recall the definition

$$\rho_{ij} = \mathbb{P}(T_j < \infty \mid X_0 = i).$$

This is the probability that the Markov Chain ever visits j . Let

$$\mathbf{B} := (\rho_{i,j})_{\substack{d'+1 \leq i \leq d \\ 1 \leq j \leq d'}} \in \mathbb{R}^{(d-d') \times d'}.$$

Find an expression for $\mathbf{B}$ in terms of $\mathbf{R}$ and $\mathbf{Q}$ given in (10.1)

Exercise 10.1.4. In the same setting of the previous exercise, find an expression for $\mathbf{P}^n$ in terms of $\mathbf{Q}$ and $\mathbf{R}$. Show that the limit of $\mathbf{P}^n$ exists as $n \rightarrow \infty$ and express this limit in terms of $\mathbf{Q}$ and $\mathbf{R}$.